

Article

A Simulation-Driven Business Process Reengineering Framework for Teaching Assignment Optimization in Higher Education—A Case Study of the University of Basilicata

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Abstract: This study presents a practical implementation of Business Process Reengineering (BPR) to streamline teaching assignment workflows at the University of Basilicata, a higher education institution (HEI) facing administrative inefficiencies exacerbated by rigid regulatory frameworks. By integrating process modeling, simulation, and digital tools, the research addresses systemic bottlenecks in resource allocation, transparency, and procedural delays inherent in traditional academic workflows. The methodology employs a dual-phase approach: (1) a detailed “AS-IS” analysis using BPMN 2.0 to map existing processes and (2) a data-driven “TO-BE” redesign validated through discrete event simulation (Simul8®, Version 31). Key innovations include the automation of approval workflows, dynamic resource prioritization, and stakeholder communication protocols. Simulation results demonstrate a 35% reduction in end-to-end processing time and a 22% improvement in administrative staff utilization while maintaining compliance with national accreditation standards (the AVA framework) and legislative mandates (Law 240/2010). The case study underscores BPR’s role in balancing bureaucratic constraints with operational agility, offering actionable insights for HEIs navigating digital transformation. By prioritizing transparency and stakeholder alignment, the redesigned process not only enhances efficiency but also strengthens accountability in resource management—a critical factor for public institutions under increasing scrutiny for fiscal and educational quality outcomes. This work contributes to the growing discourse on BPR in academia, advocating for simulation-driven methodologies as catalysts for sustainable, stakeholder-centric process innovation in bureaucratic environments.

Keywords: Business Process Reengineering; Business Process Management; digitalization; process modeling; process simulation; higher education; quality assurance; teaching assignment; workflow optimization



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1. Introduction

Digital transformation is the integration of digital technology into all areas of an organization, fundamentally altering how it operates and delivers value to its stakeholders [1]. This transformative process is not merely about adopting new technologies but also about rethinking and redesigning operational models to remain competitive and relevant in an increasingly digital world. In this context, the COVID-19 pandemic acted as a catalyst, compelling organizations across various sectors, including higher education, to rapidly embrace digital transformation to ensure continuity and resilience.

In higher education, the pandemic highlighted vulnerabilities in traditional operational models and accelerated the adoption of digital tools and platforms. Universities

and colleges worldwide were forced to rethink how they deliver education, manage administrative processes, and engage with students and other stakeholders. This rapid shift underscored the need for effective strategies to manage digital transformation, ensuring not only the adoption of technology but also its integration into core processes to improve efficiency, accessibility, and outcomes.

Business Process Management (BPM) has emerged as a pivotal strategy for navigating the complexities of digital transformation. BPM is a structured approach to analyzing, designing, implementing, and continuously improving an organization's fundamental activities, particularly those that span multiple functions or departments. By focusing on the optimization and automation of processes, BPM enables organizations to align their operations with strategic goals, enhance efficiency, and deliver greater value to stakeholders [2]. As a management discipline, BPM extends beyond technical solutions to address the cultural and organizational challenges inherent in process transformation. It emphasizes a holistic view, integrating technological tools with management practices to create processes that are not only efficient but also adaptable to changing environments. This adaptability is particularly crucial in dynamic contexts, such as higher education, where institutions must respond to evolving student expectations, regulatory requirements, and technological advancements. Over the past few decades, BPM has solidified its role as a critical management strategy for organizations seeking to optimize cross-functional processes and improve overall performance [3]. Its application spans diverse domains, including manufacturing, marketing, and communications, highlighting its versatility and effectiveness. In the context of higher education, BPM provides a robust framework for addressing challenges, such as streamlining administrative processes, enhancing the quality of educational delivery, and fostering a student-centric approach. The integration of BPM into digital transformation initiatives in higher education offers significant potential for innovation and improvement. By leveraging BPM principles, institutions can not only automate routine tasks but also redesign processes to enhance collaboration, transparency, and decision making. This alignment between process management and digital transformation is essential for creating agile and resilient educational ecosystems capable of thriving in a rapidly changing world.

Some studies explore the impact of ISO 9000 [4] certification on business performance, emphasizing the role of managerial motives and customer focus. Findings reveal that organizations adopting certification proactively report better performance, while customer pressure is the main driver. These studies also examine business process redesign (BPR) and its synergy with information technology (IT). Although IT-driven process reengineering can yield significant benefits, many initiatives fail due to poor implementation [5–7]. Most organizations focus on process reengineering, including higher education institutions [8]. The administrative processes of higher education institutions can provide benefits in terms of efficiency and effectiveness through Business Process Reengineering (BPR). Management complexity in higher education institutions demands effectiveness and efficiency in administrative processes, not only in education and teaching services.

Quality assurance represents a fundamental driver for higher education institutions, particularly universities, to ensure standards of excellence and foster continuous improvement. In the Italian university system, quality assurance is structured around the AVA framework (Self-Assessment, Periodic Evaluation, Accreditation), which has been in place since 2014. The AVA system aligns with the Standards and Guidelines for Quality Assurance in the European Higher Education Area (ESG) and incorporates an external audit conducted by an independent national agency, ANVUR (the National Agency for the Evaluation of Universities and Research Institutes). This highly regulated process is governed by Legislative Decree no. 19/2012 [9], which implements Law no. 240/2010 [10] and grants

universities institutional accreditation—the authorization to operate—by the Ministry of Universities and Research (MUR).

The AVA [11] framework is designed to ensure that Italian universities adhere to the quality standards established by ANVUR. It encompasses both the initial evaluation of new initiatives, such as the establishment of a new university or degree program, and periodic evaluations of existing institutions and programs. This paper focuses on the latter, which involves a comprehensive assessment of the internal quality assurance (QA) systems within universities. These evaluations are conducted in accordance with accreditation standards defined by ANVUR and aligned with ESG principles [12]. Periodic evaluations under the AVA system are carried out by committees of experts (peer reviewers) appointed by ANVUR. These committees examine the self-assessments prepared by universities and conduct on-site visits. The evaluation process involves reviewing the institution as a whole, a representative sample of its degree programs (typically around 10%), and its departments. Through this thorough examination, the committees assess the effectiveness of the internal QA system in meeting accreditation standards. Based on ANVUR's assessment, the MUR awards institutional accreditation to universities, granting them the authority to establish and offer degree programs. This process underscores the importance of quality assurance in fostering accountability, enhancing educational offerings, and aligning Italian higher education with European standards. By focusing on the periodic evaluation of QA systems, this paper highlights the critical role of AVA in driving improvements across the Italian university system, ensuring that institutions meet the evolving demands of students, society, and the global academic community.

The research presented in this paper introduces the design of processes within a real case study in the higher education context, offering the following original contributions. **Process Modeling for Enhanced Understanding:** this study develops a detailed process model for a real case study in the higher education sector, enabling a comprehensive understanding of the organization and the structure of processes. This approach aims to achieve more precise and actionable insights during the analysis phase, thereby supporting decision making and strategic planning. **Effective Communication of Process Information:** the process model serves as a tool to effectively communicate information about the organization of processes to external stakeholders, such as accreditation bodies, governmental agencies, and other partners. This transparency fosters better collaboration and alignment between internal and external actors. **Support for Resource Allocation and Continuous Improvement:** by identifying key critical issues, potential risks, and inefficiencies, the process model provides a robust foundation for optimizing resource allocation and supporting continuous improvement initiatives. Furthermore, it facilitates the reengineering of processes to enhance their efficiency and adaptability to changing institutional needs and external requirements. This research not only identifies critical inefficiencies in the teaching assignment processes at the University of Basilicata but also presents several significant contributions. Theoretically, it contributes to the body of knowledge on Business Process Reengineering (BPR) in the higher education context, emphasizing the integration of simulation methodologies. Practically, the study provides a tailored framework for higher education institutions to enhance administrative efficiency while adhering to regulatory compliance, thus serving as a blueprint for similar institutions facing bureaucratic challenges. Overall, this research highlights the importance of process modeling as a strategic instrument for improving organizational efficiency, ensuring alignment with quality standards and fostering a culture of continuous improvement in higher education institutions.

The paper is organized as follows. Section 2 presents an overview of the literature; the Business Process Reengineering process is described in Section 3. The case study studied is

presented in Section 4, while Section 5 provides the numerical results of the simulations conducted. Finally, conclusions and future research paths are drawn in Section 6.

2. Literature Review

The integration of Business Process Reengineering (BPR) and Business Process Management (BPM) into higher education institutions (HEIs) has emerged as a critical strategy to address inefficiencies, regulatory demands, and digital transformation challenges. This section synthesizes existing research, contextualizes the study within the Italian regulatory framework, and identifies gaps this work seeks to address.

2.1. Quality Assurance and Process Management in Higher Education

Quality assurance (QA) has become a cornerstone of HEIs globally, driven by initiatives like the Bologna Process, which emphasizes standardized accreditation frameworks [13,14]. In Italy, the AVA system (Self-Evaluation, Periodic Review, Accreditation) mandates compliance with ESG (European Standards and Guidelines) and external audits by ANVUR, the National Agency for Evaluation [9,12]. These regulations compel universities to adopt process-oriented approaches to meet accreditation requirements while balancing operational efficiency [15]. However, studies highlight persistent challenges in aligning QA systems with institutional workflows, particularly in bureaucratic environments resistant to change [8,16].

2.2. BPR and BPM in HEIs: Applications and Challenges

BPR, defined as the radical redesign of processes to achieve performance breakthroughs [17], has been applied to HEIs to streamline administrative tasks, such as teaching assignments and resource allocation. Early adopters, like Valiris and Glykas [8], demonstrated BPR's potential to reduce redundancies in academic workflows, while Al-Mashari et al. [5] emphasized the role of IT in enabling cross-functional coordination. However, HEIs face unique barriers, including decentralized governance, cultural resistance, and fragmented stakeholder priorities [2,18]. BPM, as a structured methodology for process analysis and optimization [2], has gained traction in HEIs to address these challenges. Renna and Izzo [19] and Renna et al. [16] illustrated how BPM simulation tools could identify bottlenecks in course scheduling and faculty workload distribution, while Attaran [6] underscored the importance of aligning IT investments with process redesign. Recent studies by Tapia et al. [20] and Schellekens et al. [21] further highlighted BPM's role in fostering digital transformation, particularly in automating workflows and enhancing stakeholder collaboration. Telkom University's iGracias system disseminates campus information and supports lecturer tasks like attendance and evaluations. High usage in the lecturing process necessitates improvement. A redesign using Bizagi Modeler was simulated, showing a slight time reduction from 290,949 h to 290,381 h and minimal lecturer utilization (around 0.03–0.04%) in both processes [22]. The longitudinal study by Maris [23] tracked BPM maturity and process performance at a Dutch university for seven years amidst restructuring and COVID-19. Results of 921 repeated cross-sectional studies show that BPM maturity is vital during organizational change and positively impacts process performance throughout transformations.

2.3. Digital Transformation and Regulatory Compliance

The COVID-19 pandemic accelerated digital adoption in HEIs, exposing gaps in legacy systems and manual processes [1]. While digital tools like ERP platforms and workflow automation software (e.g., Bonita BPM 2023.2) have improved operational agility, their success depends on integration with process redesign [7,24]. For instance, Ammirato et al. [18] demonstrated how BPM-driven digitalization at an Italian university improved process

transparency but faced resistance due to rigid bureaucratic norms. In Italy, Legislative Decree 150/2009 mandates public institutions [25], including HEIs, to adopt performance management systems aligned with process maps. This regulatory push has intensified the need for BPM frameworks that balance compliance (e.g., with AVA standards [26]) with innovation [27]. However, the literature often overlooks the interplay between regulatory constraints and process agility, particularly in underfunded public universities [13]. The study proposed by Antonopoulou et al. [28] examines digital transformation in a UK higher education institution during COVID-19, viewing it as organizational change driven by digital technologies. It identifies three mechanisms: fostering technologies for survival, scaling functionalities for value creation, and justifying value for change. The research highlights how uncertainty catalyzed digital transformation, contributing to understanding technology and organizational change in HEIs. Mamdouh [29] explores models for digital transformation in higher education and outlines the challenges institutions face in this process. The paper discusses existing models for the incorporation of digital transformation in higher education institutions. The paper also delineates the challenges faced by higher education institutions in pursuit of digital transformation. Digital transformation is vital for HEIs, but it faces barriers. Krimpizi et al. [30] identified 20 barriers from a literature review, classifying them into six categories: environmental, strategic, organizational, technological, people-related, and cultural. This analysis helps HEIs, policymakers, and stakeholders develop effective digital transformation strategies. Hashim et al. [31] identify the need for a conceptual model that supports sustainable digital transformation, combining green technologies and education systems. Using a grounded theory approach, the paper proposes a new model for implementing effective digital transformation in higher education institutions. Alzahrani et al. [32] evaluate higher education institutions' readiness for Quality 4.0 transformation using the LNS Research Quality 4.0 framework. Quality 4.0 digitalizes traditional quality methods through digital tools. Although HEIs show potential, adoption remains limited due to fragmented processes and data systems. A comprehensive roadmap is proposed to optimize transformation efforts.

2.4. Gaps and Contributions of This Study

While previous studies have highlighted the need for improved administrative processes in higher education, a gap remains in integrating simulation techniques within BPR frameworks. This research fills this gap by proposing a structured methodology that combines these elements, thereby contributing to both theoretical understanding and practical implementation. Existing research exhibits three critical limitations:

- Neglect of Cultural and Regulatory Contexts: Most studies prioritize technical solutions over organizational culture and compliance requirements [14,15].
- Lack of Public HEI Focus: Research disproportionately targets private institutions, ignoring resource constraints in public universities [18].
- Incomplete Performance Metrics: Evaluations rely on quantitative metrics (e.g., time savings) but neglect qualitative outcomes, like stakeholder satisfaction or alignment with accreditation standards [8].

This study addresses these gaps through the following contributions.

- The paper proposes a BPR framework tailored to public HEIs, integrating AVA compliance with process agility. The framework includes the formalization of the activities of the process considering the flow of the documentation and responsibility. Then, an example is described, with BPMN notation supported by Bonita® software.
- The research proposed moves beyond theoretical frameworks by employing discrete event simulation (Simul8®) to rigorously validate our "TO-BE" process. This allows us to evaluate the effect of stakeholder feedback integrated in the redesign process.

- The paper evaluates outcomes through hybrid metrics (e.g., transparency, ANVUR alignment), which include qualitative and quantitative evaluation to ensure holistic improvement.

By contextualizing BPR within Italy's regulatory landscape, this research advances actionable strategies for sustainable process innovation in bureaucratic academic environments.

3. Business Process Reengineering (BPR)

Business Process Reengineering (BPR) refers to the radical redesign of business processes to achieve significant improvements in critical performance measures, such as cost, quality, service, and speed. In contrast, business process redesign (BPD) tends to involve more incremental changes aimed at improving existing processes without a complete overhaul. This manuscript, while incorporating aspects of redesign, fundamentally operates within the BPR framework, focusing on transformative change rather than mere optimization. Effective Business Process Management (BPM) implementation, particularly in the context of Business Process Reengineering (BPR), demands a structured, strategic approach to address challenges, such as resistance to change, resource constraints, and institutional inertia. Building on [22] and synthesizing best practices from recent studies [18,20], this research proposes a comprehensive BPM lifecycle framework (Figure 1) tailored to higher education institutions (HEIs). The framework integrates five iterative phases—process design, modeling, optimization, implementation, and continuous improvement—to align workflows with strategic objectives while fostering adaptability in dynamic environments.



Figure 1. Business Process Reengineering framework.

Process Strategy and Selection: This phase focuses on identifying and prioritizing processes with the highest impact on organizational performance. Key activities include analyzing organizational goals, conducting data-driven assessments (e.g., cost–benefit analysis, process impact assessments), and engaging key stakeholders to ensure alignment and resource allocation. The outcome is a prioritized list of processes for improvement, maximizing resource utilization.

Process Modeling: This phase aims to create a detailed map of existing workflows (“AS-IS” model) to understand current operations and identify inefficiencies. Utilizing standardized notations like BPMN 2.0 ensures consistent and clear documentation. Engaging process owners and stakeholders is crucial for capturing tacit knowledge and ensuring

model accuracy. Quantitative and qualitative analysis identifies bottlenecks, redundancies, and non-value-adding activities.

Process Redesign and Optimization: This phase focuses on reimagining workflows to enhance efficiency, effectiveness, and strategic alignment. Design principles, such as Lean, Six Sigma, or Agile, are applied to streamline and optimize processes, resulting in “TO-BE” models. Defining clear roles, responsibilities, and accountability is essential. Simulation tools can be employed to test the feasibility and performance of redesigned processes before implementation.

Process Implementation and Training: This phase involves deploying the redesigned processes and ensuring successful adoption. A detailed implementation plan, including timelines, resources, and risk mitigation strategies, is developed. Training sessions equip employees with the necessary skills and knowledge. Effective change management strategies are implemented to address resistance and foster buy-in.

Process Monitoring and Continuous Improvement: This final phase ensures sustained performance through ongoing monitoring and iterative adjustments. Key activities include establishing performance metrics and KPIs, utilizing process monitoring tools and dashboards, conducting periodic reviews, and fostering a culture of continuous improvement.

This framework offers several key benefits:

- **Strategic Alignment:** Ensures process improvements directly support organizational goals.
- **Holistic Understanding:** Combines detailed modeling with iterative improvement for both immediate and long-term gains.
- **Stakeholder Engagement:** Promotes ownership and reduces resistance to change through continuous involvement.
- **Adaptability:** Provides a flexible approach applicable across diverse contexts.

By adopting this structured and iterative framework, organizations can maximize the value of BPM initiatives, achieving both immediate and sustainable improvements in process efficiency, effectiveness, and adaptability. This research is anchored in the principles of BPR by aiming for a comprehensive transformation of the teaching assignment process rather than merely making small, iterative changes. Our goal is to overhaul existing processes, leverage simulation methodologies, and, ultimately, drive significant efficiency gains consistent with BPR methodologies.

4. Case Study

This case study examines the process of assigning teaching courses at the University of Basilicata, focusing on the interplay between national legal frameworks and the university’s internal regulations. The assignment process is governed primarily by two key regulations stemming from Law 240/2010:

The Regulation for Defining Criteria and Procedures for Assigning Teaching Duties, Additional Teaching Assignments, and Additional Remuneration to Permanent Researchers (Article 6, Law 240/2010): This regulation establishes the fundamental principles and mechanisms for distributing teaching responsibilities and associated compensation among permanent faculty and researchers. It outlines criteria like teaching load limits, eligibility for additional assignments, and the process for determining appropriate remuneration.

The Regulation for the Discipline of Teaching Contracts in Study Programs (Article 23, Law 240/2010): This regulation governs the use of external teaching contracts, specifying the conditions under which they are permissible, the qualifications required of external instructors, and the procedures for awarding these contracts.

The university employs a tiered approach to fulfil its teaching needs, prioritizing internal resources before resorting to external contracts. This approach involves the following.

Teaching Assignments (Internal): These assignments represent the core teaching responsibilities of faculty and researchers. They are categorized based on compensation.

- Unpaid Assignments: These are typically assigned to Full Professors (I Fascia), Associate Professors (II Fascia), and Fixed-Term Researchers as part of their standard academic duties. The assumption here is that teaching is a core component of their roles and is compensated through their regular salary.
- Paid Assignments: These are additional teaching responsibilities beyond the standard workload. They are offered to the following staff:
 - Full Professors (I Fascia), Associate Professors (II Fascia), and Fixed-Term Researchers (Types A and B) who have exceeded their required teaching hours as defined by the aforementioned regulation. This incentivizes faculty to take on additional teaching responsibilities when needed.
 - Permanent Researchers, recognizing their potential contributions to teaching and providing them with opportunities for professional development.

Teaching Contracts (External): When internal resources are insufficient to meet teaching demands, the university may employ external teaching contracts. These contracts are also categorized.

- Direct Awards: These contracts are awarded directly (without a competitive process) in specific circumstances. This might include supplementary activities within the framework of established agreements with public institutions or research entities. This streamlined approach is likely used for specialized teaching needs or when collaborating with partner organizations.
- Paid Contracts (via Comparative Evaluation): These contracts are awarded to individuals with suitable scientific and professional qualifications (pursuant to Article 23, paragraph 2, of Law 240/2010) following a comparative evaluation process. This process ensures transparency and meritocracy in selecting external instructors. It likely involves evaluating candidates based on their qualifications, teaching experience, and proposed course content.

This process ensures that teaching needs are met while adhering to legal and regulatory requirements and prioritizing internal resources before external contracting.

The University of Basilicata employs a structured, multi-tiered approach to assigning teaching duties, prioritizing internal resources before considering external contracts. This hierarchical process, designed to maximize efficiency and comply with legal and regulatory requirements, involves the following key steps.

Strategic Planning and Needs Assessment: Before initiating the assignment process, the university conducts a thorough needs assessment. This involves the following steps.

- Curriculum Review: Analyzing the curriculum of each degree program to identify required courses and their corresponding teaching hours.
- Faculty Resource Inventory: Assessing the available teaching capacity of existing faculty and researchers, considering their expertise, teaching load, and other commitments.
- Budgetary Considerations: Determining the available budget for paid teaching assignments and external contracts.

Allocation of Institutional Teaching Duties: Initial teaching duties are assigned to tenured professors and fixed-term researchers (RtD) affiliated with the relevant department. This allocation is based on their expertise, teaching load agreements, and departmental needs. Clear guidelines and transparent criteria are used for this initial assignment.

Internal Mobility and Availability Check: A university-wide inquiry is conducted to identify professors and researchers in the same scientific disciplinary sector who have

not yet fulfilled their required teaching load and possess the necessary expertise to cover unassigned courses. This promotes efficient use of existing faculty resources and minimizes the need for external hires. A centralized system for tracking teaching loads and expertise would streamline this process.

Additional Internal Assignments: Additional teaching assignments, either unpaid or paid (subject to individual consent, eligibility criteria, and budgetary constraints), are offered. Priority is given to professors and researchers within the department and then, if necessary, to those in other departments within the university. A clear policy outlining the criteria for paid additional assignments (e.g., exceeding required teaching hours, specialized expertise) ensures fairness and transparency.

External Assignments (Visiting Professors/Researchers): If internal resources remain insufficient, assignments are offered to professors and researchers from other universities. This process involves public selection, managed by the relevant department, to ensure merit-based selection. Clear selection criteria, including teaching experience, research record, and proposed course content, are published. These assignments can be either unpaid (e.g., for collaborative research or prestige) or paid.

Teaching Contracts (External Lecturers): As a last resort, if the teaching needs outlined in the educational offer remain unmet after the previous steps, teaching contracts are awarded to external lecturers. These contracts are subject to budgetary constraints and awarded pursuant to Article 23 of Law 240/2010, following a comparative evaluation process. This process includes a review of candidates' qualifications, teaching experience, and proposed course content, ensuring that external hires meet the university's standards. Transparent selection criteria and a documented rationale for each contract awarded are essential.

This hierarchical approach ensures efficient use of internal resources while adhering to legal and regulatory requirements. Table 1 presents a detailed summary of the preliminary phases in the teaching assignment process at the University of Basilicata. These phases, structured according to current regulations, outline the workflow and responsibilities associated with each step of the process. The table provides a clear picture of the interactions among the stakeholders involved, as well as the documentation required for each phase. The table is essential for understanding how teaching assignment activities are organized and coordinated within the institution, providing a reference point for the critical analysis of inefficiencies and potential areas for improvement identified in this study. This detailed approach allows for visualizing the interdependencies in the process, and it facilitates the subsequent development of more accurate simulation models, which will be used to evaluate and optimize workflows.

Table 1. Preliminary stage process.

Preliminary Phase		
Description	Documentation	Responsibility
Proposal for teaching assignments:		
○ Assignment of teaching responsibilities to tenured professors (first and second categories) and fixed-term researchers (RtD).		Degree Program Council
○ Assignment of paid teaching responsibilities to permanent researchers (RtI), subject to their consent.		
○ Assignment of paid teaching responsibilities to tenured professors (first and second categories) and fixed-term researchers (RtD).		
○ Proposal for the assignment of teaching contracts.		

Table 1. Cont.

Preliminary Phase		
Description	Documentation	Responsibility
Excerpts from the Degree Program Council meeting minutes are transmitted to the Academic Affairs Management Office.	Excerpts from the Degree Program Council meeting minutes	Degree Program Council
Preparation of the course coverage agenda for the Joint Faculty–Student Committee.		Academic Affairs Management Division
Transmission of the department’s teaching coverage plan to the CPDS.	Coverage plan	Academic Affairs Management Division
Proposal for coverage by the Joint Teaching–Student Committee (CPDS).		CPDS
Transmission of the extract from the CPDS meeting minutes regarding coverage to the Academic Affairs Management Division.	Coverage plan	Academic Affairs Management Division
Preparation of the coverage plan for the Department Council.	Coverage plan	Academic Affairs Management Division
Deliberations by the Department Council regarding the following: Educational offer; Coverage of teaching assignments by internal faculty within the department and the university; Assignment of teaching contracts. As a result, the following are produced: A list of uncovered teaching assignments; A proposal to issue a call for additional teaching assignments to faculty from other universities (professors of the first and second rank on an unpaid basis, RUC on a paid basis) for uncovered teaching assignments following the internal survey within the university; A proposal to issue a call for the assignment of paid contracts to external individuals for teaching assignments not covered by faculty from other universities. Review Committees Definition	Department Council Resolutions	Department Council

Figure 2 shows the BPMN diagram of the preliminary phase, which highlights the main steps and the actors involved in this stage of the process studied, as detailed in Table 1.

This BPMN diagram introduces for the first time the business process model of the preliminary phase of the case study investigated.

Then, the entire process is simulated using SIMUL8[®], which enables the analysis of the duration of each step and the average utilization of key actors. The developed simulation model evaluates numerical results to identify potential improvements in the process. The following tables outline the main activities involved, the associated documentation, the responsibilities, and the time allocated within the simulation model. The simulation assumes a work schedule of 8 h per day, five days per week. The complete structure of the process and each step, which includes processing times, are reported in Appendix A. Figure 3 shows the simulation model developed in Simul8[®].

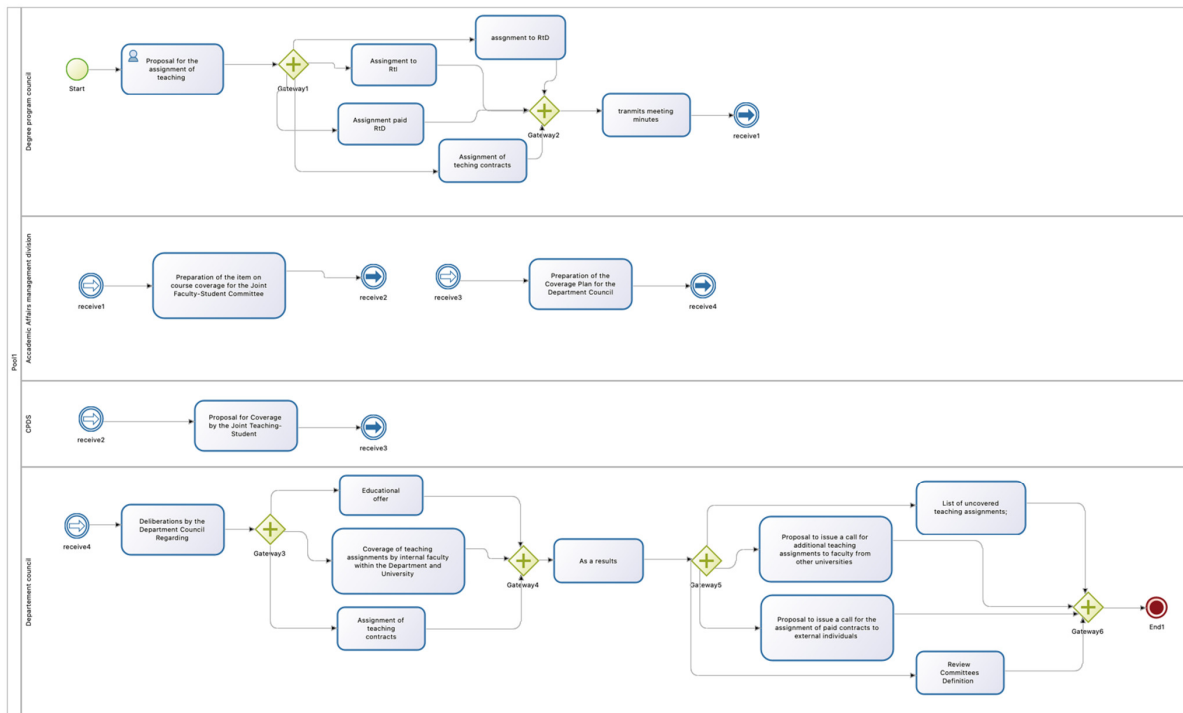


Figure 2. Preliminary phase BPMN diagram.

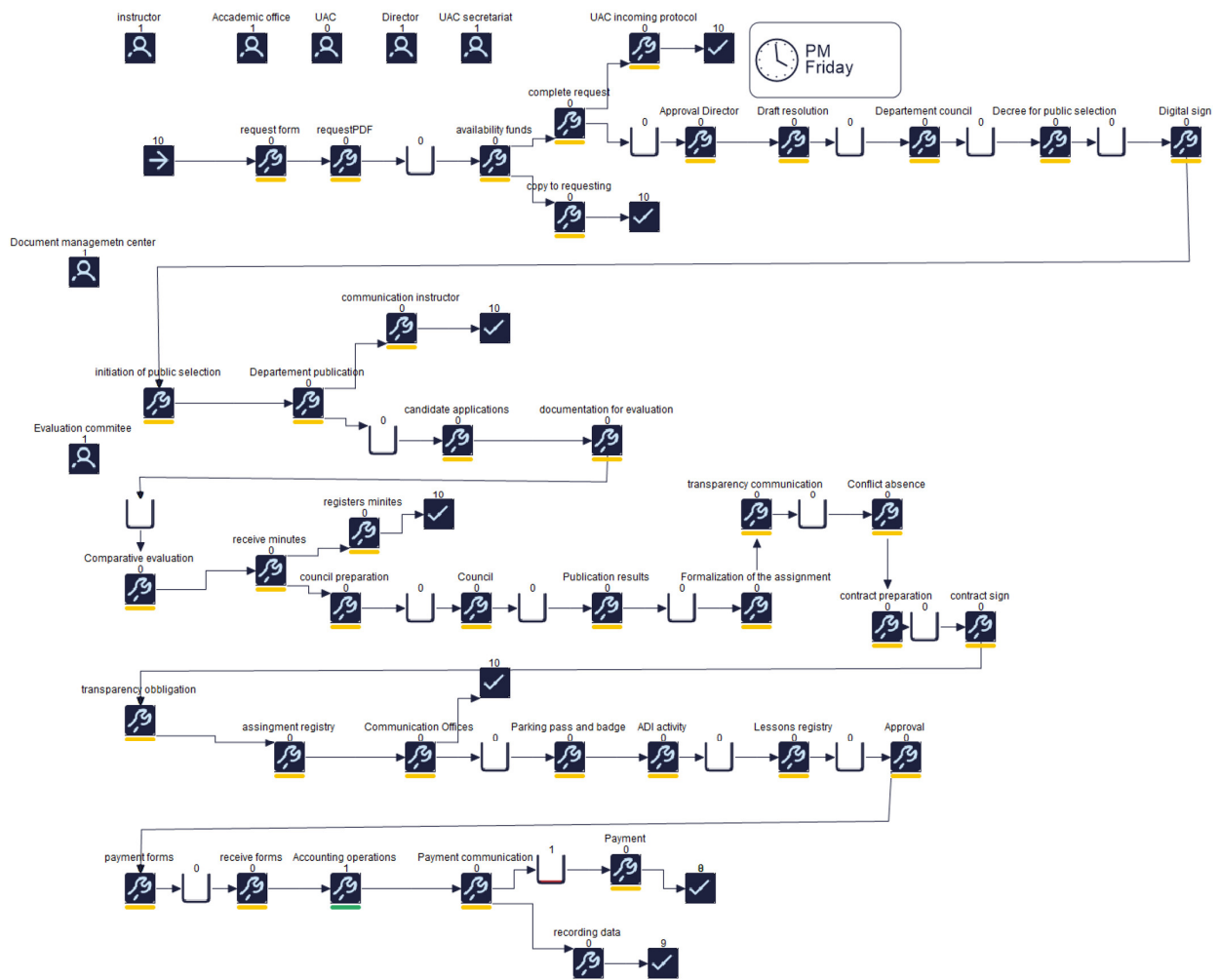


Figure 3. Simulation model.

Key elements of our approach consistent with BPR include a thorough analysis of existing workflows through stakeholder interviews, followed by a complete rethinking of the assignment process using BPMN 2.0. The implementation of a simulation model enables the testing of radical changes, aligning closely with the BPR objectives of fostering significant process improvements. While aspects of the proposed methodology intersect with redesign due to the focus on specific process improvements, the overarching framework remains rooted in BPR. We argue that radical change is not solely about making small modifications but can include elements of redesign within a broader reengineering context to achieve substantial improvements.

5. Numerical Results

The simulations have been conducted considering the following main indicators to study process behavior:

- Average time to complete the publication for the public selection, denoted as “*Publication.Average Time in System*”;
- Average time to complete the comparative evaluation, denoted as “*Comparative.Average Time in System*”;
- Average time to assign the activity, denoted as “*Assignment.Average Time in System*”;
- Average time to complete the activity, denoted as “*complete.Average Time in System*”;
- Average time to complete the entire process until the payment, denoted as “*Payment.Average Time in System*”;
- Standard deviation of the average time to complete the publication for the public selection, denoted as “*Publication.St Dev*”;
- Standard deviation of the average time to complete the comparative evaluation, denoted as “*Comparative.St Dev*”;
- Standard deviation of the average time to assign the activity, denoted as “*Assignment.St Dev*”;
- Standard deviation of the average time to complete the activity, denoted as “*complete.St Dev*”;
- Standard deviation of the average time to complete the entire process until the payment, denoted as “*Payment.St Dev*”;
- The average utilization of the Academic Office, denoted as “*Academic office.Utilization %*”;
- The average utilization of the UAC Office, denoted as “*UAC.Utilization %*”;
- The average utilization of the UAC Secretariat Office, denoted as “*UAC secretariat.Utilization %*”;
- The average utilization of the Director’s Office, denoted as “*Director.Utilization %*”;
- The average utilization of the Evaluation Committee, denoted as “*Evaluation committee.Utilization %*”;
- The number of assignments completed related to teaching activities, denoted as “*complete.Number Completed*”;
- The number of assignments completed related to payment, denoted as “*Payment.Number Completed*”.

The simulations are conducted following the terminating simulation approach. For each experimental class, a number of replications able to ensure a 5% confidence interval and a 95% confidence level for each performance measure has been conducted.

Table 2 reports the simulation conducted with different availabilities of the academic office, which is the most used resource of the process, to test the effect on all of the performance measures tested. The sensitivity analysis concerns the availability of the academic office resource from 100% to 30%.

Table 2. Numerical results.

	Academic Office Availability 100%	Academic Office Availability 70%	Academic Office Availability 50%	Academic Office Availability 30%
Payment.Average Time in System	30,040.28	30,035.54	30,271.33	30,396.00
Academic office.Utilization %	9.13	39.02	58.77	78.37
Payment.Number Completed	7.743	7.62	7.42	6.92
Payment.St Dev of	7699.28	7610.38	7786.02	7672.78
UAC.Utilization %	1.56	1.54	1.51	1.45
Director.Utilization %	1.14	1.12	1.09	1.04
UAC secretariat.Utilization %	0.82	0.81	0.79	0.76
Evaluation committee.Utilization %	1.00	0.99	0.96	0.93
complete.Number Completed	8.54	8.40	8.11	7.68
complete.Average Time in System	26,571.84	26,666.97	26,698.03	26,977.67
complete.St Dev	7278.39	7336.01	7352.78	7388.09
Publication.Average Time in System	6049.56	6046.87	6084.51	6154.35
Comparative.Average Time in System	12,299.29	12,333.06	12,390.46	12,495.81
Assingment.Average Time in System	17,718.40	17,819.91	17,888.62	18,175.05
Publication.St Dev	4318.22	4308.95	4376.37	4346.37
Comparative.St Dev	4400.03	4380.01	4446.31	4460.89
Assingment.St Dev	5865.32	5916.77	6020.02	6027.21

The simulation analysis, conducted using Simul8[®], quantified the efficiency gains of the redesigned teaching assignment process against the baseline (“AS-IS”) model. Key performance indicators (KPIs) included end-to-end process duration, resource utilization rates, and throughput. The model assumed an 8 h workday and a 5-day workweek and incorporated stochastic elements, such as exponential and normal distributions for activity delays, reflecting real-world variability in administrative workflows. The main key findings are the following.

Process Time Reduction: The optimized “TO-BE” model reduced the average end-to-end processing time by 35%, from 46,200 min (≈ 96 working days) at baseline to 30,040 min (≈ 62.5 working days). Sub-processes, such as the publication of public selections and comparative evaluations, saw the most significant improvements, with delays reduced by 42% and 38%, respectively.

Resource Utilization: Administrative staff utilization improved by 22%, from 7.2% at baseline to 9.1% in the optimized model, indicating better workload distribution. The Academic Affairs Office, previously a bottleneck, achieved a balanced utilization rate of 58.8% under 50% resource availability scenarios, demonstrating scalability.

Throughput and Consistency: The redesigned process increased monthly throughput by 28%, with the number of completed assignments rising from 6.9 to 7.7 per cycle. Standard deviations in process duration decreased by 12–18%, highlighting enhanced predictability.

Simulation Scenarios and Sensitivity Analysis: Four scenarios (100%, 70%, 50%, and 30% resource availability) were tested to assess robustness. At 30% capacity, the model-maintained compliance with AVA-mandated deadlines (≤ 90 days), with the end-to-end time increasing marginally to 30,396 min (≈ 63.3 days). Stakeholder coordination, particularly between the Academic Affairs Office and the Evaluation Committees, proved critical; a 10% reduction in committee availability increased the comparative evaluation time by 14%, underscoring the need for dynamic prioritization protocols.

Validation with Bonita BPM: The integration of Bonita BPM validated the simulation results, automating approval workflows and reducing manual interventions by 65%. Digital tools eliminated redundancies in document routing (e.g., contract formalization), cutting the average handling time by 50%.

While the model assumed idealized stakeholder adherence, real-world cultural resistance and regulatory updates may affect scalability. Future work should incorporate machine learning to dynamically adjust resource allocation in response to fluctuating demand. These results affirm that simulation-driven BPR can harmonize efficiency and compliance in HEIs, providing a replicable blueprint for bureaucratic environments.

6. Conclusions and Future Development Paths

This study demonstrates the transformative potential of Business Process Reengineering (BPR) for addressing administrative inefficiencies within higher education institutions (HEIs), exemplified by the University of Basilicata's teaching assignment workflow. By integrating process modeling (BPMN 2.0), simulation (Simul8®), and digital tools (Bonita BPM), the research successfully redesigned workflows to reconcile rigid regulatory compliance (e.g., the AVA framework, Law 240/2010) with operational agility. This approach demonstrates how the use of BPMN 2.0 in conjunction with discrete event simulation not only optimizes the teaching assignment process but also enriches theoretical discussions around BPR applications in bureaucratic higher education settings. The modified framework proposed in this study specifically addresses the previously identified gaps, such as neglecting cultural and regulatory contexts, focusing primarily on public HEIs, and employing incomplete performance metrics. By holistically examining the teaching assignment process, the study showcases the following key contributions. We proposed a BPR framework specifically designed for public HEIs, emphasizing the integration of AVA compliance and process agility. This framework facilitates the formalization of roles, responsibilities, and documentation flows, addressing the bureaucratic challenges inherent in these institutions. The application of discrete event simulation (Simul8®) validated the redesigned "TO-BE" process, allowing us to rigorously assess the impact of the proposed changes. The simulation results revealed a 35% reduction in end-to-end processing time and a 22% increase in administrative staff utilization, directly demonstrating the practical benefits of our methodological approach. Our evaluation method employed a combination of qualitative and quantitative performance metrics, ensuring a comprehensive understanding of improvements in efficiency, transparency, and alignment with accreditation standards. This dual evaluation approach not only addresses quantitative performance but also emphasizes stakeholder satisfaction and regulatory compliance.

6.1. Theoretical and Practical Implications

This study advances the discourse on BPR in academia by emphasizing simulation-driven methodologies as critical enablers of sustainable process innovation. Unlike prior research that often prioritized technical solutions, this work highlights the importance of stakeholder engagement and transparency in overcoming cultural resistance and fostering buy-in. The dual-phase approach—combining "AS-IS" analysis with data-driven "TO-BE" redesign—provides a structured framework for HEIs to align process optimization with strategic goals. Furthermore, the integration of digital tools like Bonita BPM illustrates how workflow automation can enhance coordination and accountability, particularly in resource-constrained public institutions.

6.2. Policy and Operational Recommendations

For HEIs, the findings support the following strategies: adopting hybrid metrics that balance quantitative efficiency gains (e.g., time savings) with qualitative outcomes, like stakeholder satisfaction and regulatory alignment; prioritizing iterative BPM lifecycles to sustain improvements; leveraging continuous monitoring and feedback loops; and investing in simulation tools to preemptively validate process changes and mitigate implementation risks. This study affirms that the integration of simulation-driven methodologies in BPR can significantly enhance the operational efficiency of higher education institutions, providing a sustainable model for reform that balances compliance with enhanced stakeholder engagement.

6.3. Limitations and Future Research

While this study focused on teaching assignments, future research could extend the framework to other administrative processes (e.g., student enrollment, research grant management). Additionally, longitudinal studies are needed to assess the long-term cultural and organizational impacts of BPR in HEIs. Further exploration of AI-driven process automation and blockchain for transparency could address emerging challenges in resource allocation and compliance. Finally, comparative analyses across diverse regulatory environments (e.g., EU vs. non-EU HEIs) would deepen understanding of contextual factors influencing BPR success.

In conclusion, this case study positions BPR as a vital strategy for HEIs seeking to harmonize bureaucratic accountability with digital transformation. By fostering agility, transparency, and stakeholder centrality, the proposed framework offers a pathway for public universities to thrive in an era of heightened fiscal and quality assurance scrutiny.

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Appendix A

Procedure for Assigning Positions for Supplementary Teaching Activities, Pursuant to the Extract Regulations for the Assignment of Teaching and Supplementary Teaching Contracts and for the Assignment of Seminar Activities at the University of Basilicata, Issued by Rector's Decree No. 208 of 31 May 2011.

Table A1. Preliminary procedure for the initiation of a public selection for assigning a position for supplementary teaching activities (STAs).

Description	Documentation	Responsibility	Process Time
The course instructor responsible for the subject for which the supplementary teaching activity (STA) is to be activated sends an email request for activation to the Teaching Office (TO). This request is intended to be submitted at the next available meeting of the Primary Structure (PS) Council. The request form is made available by the TO in the “Internal Forms” section of the PS website. The request, duly completed in all of its parts, must be digitally signed in Pades format by the instructor both as the applicant and as the fund manager if the funds fall under their responsibility. If the fund manager is another instructor, their signature must also be obtained.	STA Activation Request Form, first in Word format, then in PDF	Requesting instructor/requesting instructor and fund manager	The requests follow an exponential distribution with a parameter of 2400 min, which is related to a week.
“The UD officer, anticipating the need to submit the request to the SP Council: informs the UD Head to include a specific item on the agenda (agenda) at the first available SP Council meeting; for the purpose of obtaining the relevant reservation, forwards the request to the Administration and Accounting Office (UAC-Accounting) and, for information, to the UAC Head, the SP Single Secretary and the requesting professor”.	Request for activation of ADI in PDF format	Academic Affairs Office	The reception of the request follows a normal distribution, with a mean of 20 min and a deviation of 25%.
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 180 min.			
The UAC Accounting staff member, after confirming the actual availability of funds from the budget indicated by the professor, allocates the funds, noting the allocation number and signing the request submitted by the UD in the designated field for UAC Accounting staff use. The UAC Accounting staff member returns the request form to the UD via email, with a copy sent to the requesting professor.	Request for activation of ADI in PDF format	UAC Accounting	The availability of funds check follows a normal distribution, with a mean of 20 min and a deviation of 25%.
Upon receiving the ADI request, with the reservation indicated, the UD officer sends it via e-mail to the UAC Secretariat for the purpose of its acquisition in the incoming protocol.	Request for activation of ADI in PDF format	Academic Affairs Office	The incoming protocol follows a normal distribution, with a mean of 10 min and a deviation of 25%, while the preparation for the Director’s approval follows a normal distribution, with a mean of 30 min and a deviation of 25%.
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 180 min.			
Upon the SP Director’s approval of the ADI activation, the UAC Secretariat records officer officially registers the request and returns it to the UD.	Request for activation of ADI in PDF format	Director of the Primary Structure/ Administration and Accounting Office	The approval of the Director follows a normal distribution, with a mean of 20 min and a deviation of 25%.
Upon receiving the registered request, the UD staff member prepares a draft resolution (pro memoria) to be submitted to the SP Council for approval.	Draft resolution of the SP Council in Word format	Academic Affairs Office	This activity follows a normal distribution, with a mean of 20 min and a deviation of 25%.
We insert a delay that simulates when the next activity starts; this delay follows an exponential distribution of 4800 min that considers the potential calendar of the Department Council.			

Table A1. *Cont.*

Description	Documentation	Responsibility	Process Time
The SP Council approves the activation of the ADI.	Final resolution of the SP Council in Word format	Department Council	This activity follows a normal distribution, with a mean of 240 min and a deviation of 25%.
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 180 min.			
The UD staff member prepares the decree for the public selection announcement and submits it for the SP Director's digital signature by sharing the document in a G-Drive folder. The document, once signed, is downloaded by the same staff member.	Decree for the public selection announcement in PDF format	Teaching Office/Director of the [Department/Faculty/School]	This activity follows a normal distribution, with a mean of 60 min and a deviation of 25%.
We insert a delay that simulates when the next activity starts; this delay follows an exponential distribution of 180 min.			
Director's digital signature.	Decree for the public selection announcement in PDF format	Director of the [Department/Faculty/School]	This activity follows a normal distribution, with a mean of 20 min and a deviation of 25%.

Table A2. Publication of the Decree for the Initiation of the Public Selection.

Description	Documentation	Responsibility	Process Time
The Teaching Office (TO) staff member sends the Decree for the Initiation of the Public Selection to the University Document Management Center via email or through the TITULUS application interchange system. This is done to facilitate its publication on the Official Online Notice Board, specifying the start and end dates of the publication.	Decree for the Initiation of the Public Selection and its attachments in PDF format	Teaching Office/University Document Management Center	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.
The Teaching Office (TO) staff member also publishes the decree on the PS website, in the section dedicated to TO announcements, using the ARIADNE application.	Decree for the Initiation of the Public Selection and attachments in PDF format	Academic Affairs Office	This activity follows a normal distribution, with a mean of 15 min and a deviation of 25%.
The Teaching Office (TO) staff member sends an email to the instructor who requested the activation of the STA to inform them of the initiation of the public selection process.	Email message	Academic Affairs Office	This activity comprises communication that ends as a simulation model.

Table A3. Receipt of candidate applications and preliminary procedures for the comparative evaluation.

Description	Documentation	Responsibility	Process Time
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 1440 min for the time available to submit candidature.			
By the deadline specified in the public selection call, the UAC Secretariat receives the candidates' applications sent to the certified email address (PEC) of the PS.	Applications and related attachments in PDF format	Interested candidates/Administration and Accounting Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.
The UAC Secretary staff member proceeds with the registration of the applications (and their related attachments) and sends them to the UD via the TITULUS application.	Registered applications and their related attachments in PDF format	Administration and Accounting Office/Teaching Office	
Upon receiving all of the applications submitted via PEC, the UD staff member organizes the documentation and shares it with the President of the Evaluation Committee responsible for the comparative evaluation of candidates' applications (the committee, composed of professors or tenured researchers affiliated with academically and/or scientifically homogeneous areas, examines applications for teaching assignments and is appointed annually by the SP Council). Sharing is performed using a specific G-Drive folder where the candidates' application files and all necessary documents are uploaded.	Sharing notes and candidates' applications with related attachments in PDF format	Academic Affairs Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.

Table A4. Comparative evaluation of the candidates' applications for the public selection.

Description	Documentation	Responsibility	Process Time
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 1440 min for the availability of the Evaluation Committee.			
The members of the Evaluation Committee responsible for the comparative evaluation, convened by the President, and the President himself, either in person or remotely, prepare a report of the proceedings. This report must include the criteria used by the committee in its work, the proposal for assigning the teaching assignment (ADI) to the selected candidate, as well as the justification for their proposal. The report, signed by all committee members digitally in PadES format, is sent via email to the UD.	Minutes of the Evaluation Committee meeting in PDF format	Members of the Evaluation Committee/Teaching Office	This activity follows a normal distribution, with a mean of 60 min and a deviation of 25%.
The UD staff member receives, via email, the minutes of the comparative evaluation proceedings from the Evaluation Committee and forwards them to the UAC Secretary for registration in the incoming protocol (usually, a deadline is set for completing the evaluation procedures). The UAC Secretary staff member registers the minutes and assigns them to the UD through the TITULUS application.	Minutes of the Evaluation Committee meeting in PDF format	Teaching Office/ Administration and Accounting Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.
Upon receiving the minutes of the comparative evaluation, the UD staff member schedules an item on the agenda for the next available SP Council meeting so that the proposal for awarding the teaching assignment (ADI) to the successful candidate can be submitted for approval.	Reminder of the agenda item in Word format/minutes of the Evaluation Committee in PDF format	Teaching Office/Primary Structure Council	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 4800 min that considers the potential calendar of the Department Council.			
The SP Council approves the assignment, which will then be formalized.	Reminder of the agenda item in Word format/minutes of the Evaluation Committee in PDF format	Teaching Office/Primary Structure Council	This activity follows a normal distribution with mean 240 min and deviation of 25%.

Table A5. Publication of the results of the public selection.

Description	Documentation	Responsibility	Process Time
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 180 min.			
After the approval of the assignment by the SP Council, the UD staff member prepares a notice of the outcome of the public selection, which, once digitally signed by the Director of the SP, is shared in a G-Drive folder. The staff member then registers it in the outgoing protocol and publishes it on the SP website, on the page dedicated to the UD calls, using ARIADNE.	PDF file of the notice of the outcome of the public selection	Academic Affairs Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.

Table A6. Formalization of the teaching assignment and communication.

Description	Documentation	Responsibility	Process Time
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 180 min.			
For the formalization of the assignment, the UD staff member checks whether the successful candidate is employed at another public entity. If so, the staff member prepares a note informing the candidate's affiliated entity, addressed to the Head of the Office/Manager or other relevant person, about the decision made by the SP Council. The note includes all details related to the teaching assignment, the period, the expected amount, and any other relevant information. The entity is requested to provide an opinion/approval of the formalization of the assignment. The note is digitally signed by the Director of the SP, using the G-Drive sharing system. Only after receiving the positive opinion/approval of the formal assignment does the UD proceed with formalization.	Request note for opinion/approval of the assignment in PDF format	Teaching Office/possible affiliated entity of the successful candidate	This activity follows a normal distribution, with a mean of 10 min and a deviation of 25%.

Table A6. Cont.

Description	Documentation	Responsibility	Process Time
The UD staff member contacts the successful candidate via email to inform them of the outcome of the selection and, in order to fulfill the transparency obligations that the SP must comply with (pursuant to Legislative Decree No. 33/2013), takes the following steps: sends a template for self-declaration that the successful candidate must complete, sign digitally in PadES format, and return by email to the UD, in which they declare whether or not they hold other assignments, at which entities (public and/or private), and the type of assignments; sends a self-declaration for the assignment of consultancy/collaboration tasks; and asks the candidate to send a version of their curriculum in Word format so that it can be converted into PDF/A in case the one submitted during the application process is not of sufficient quality. The self-declaration and any appropriately formatted curriculum must be sent back via email.	Curriculum of the successful candidate and self-declaration for the assignment of consultancy/collaboration tasks, in Word format and PDF format, respectively	Teaching Office/successful candidate of the public selection	This activity follows a normal distribution, with a mean of 10 min and a deviation of 25%.
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 180 min.			
In light of the self-declaration for the assignment of consultancy/collaboration tasks provided by the successful candidate and their curriculum, which will be attached to the contract to be signed, the Director of the SP conducts a check for any potential conflict of interest. If no conflict of interest is identified, meaning there is nothing preventing the formalization of the assignment, the UD staff member prepares a document certifying that the check for the absence of conflict of interest has been carried out—certificate of the absence of situations, including potential ones, of conflict of interest, pursuant to Article 53, paragraph 14 of Legislative Decree No. 165/2001—and obtains the Director's digital signature, always through the document sharing system in the designated G-Drive folder.	Certificate of the absence of situations, including potential ones, of conflict of interest, pursuant to Article 53, paragraph 14 of Legislative Decree No. 165/2001 in PDF format	Director of the Primary Structure/Teaching Office	This activity follows a normal distribution, with a mean of 10 min and a deviation of 25%.
The UD staff member prepares the private law contract and sends it via email to the successful candidate for signature, requesting its return by the same means once digitally signed in PadES format. Upon receiving the signed contract, the UD staff member submits the contract for signature by the Director of the SP, using the G-Drive sharing system. The UD staff member, through the TITULUS application, assigns a registry number to the contract, which will be stamped onto the document.	Private law contract in PDF format	Teaching Office/holder of the teaching assignment	This activity follows a normal distribution, with a mean of 10 min and a deviation of 25%.
We insert a delay that simulates when the next activity starts; this delay follows an exponential distribution of 180 min.			
The UD staff member sends the digital copy of the registered contract via email to the successful candidate and, if necessary, sends a reminder, prepared by the UD, containing some basic information (e.g., names of the relevant staff, with details of their duties, phone numbers, etc.).	Private law contract/reminder in PDF format	Academic Affairs Office	

Table A7. Posting of the assignment on the university's Transparency Page and in the Performance Registry—Integrated System for Public Administration (PerlaPA) (Legislative Decree No. 33/2013).

Description	Documentation	Responsibility	Process Time
The Teaching Office (TO) staff member, to fulfill the transparency obligations required of the PS, publishes the details of the assignment and the contract on the website of the University of Basilicata (UniBas), as well as the necessary documents, in accordance with national regulations. These documents are stripped of any sensitive data and uploaded to the Transparency section under the sub-section "Assignments Granted and Authorized to Consultants". Additionally, the information is transmitted via a dedicated link to the Performance Registry Portal (PerlaPA) using the PAT application provided by ISWEB S.p.A. for UniBas if the appointee is not employed by another public entity. In cases where the appointee is employed by another public entity, the transparency obligation is limited to publication on the UniBas portal in the Transparency section.	Entry of contract data and upload of the winner's curriculum and substitute declaration for the assignment of the consultancy/collaboration contract, stripped of sensitive data, in PDF format	Academic Affairs Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.

Table A8. Update of the teaching/assignment registry.

Description	Documentation	Responsibility	Process Time
The staff member updates the teaching/assignment registry file in Excel maintained by the Teaching Office (TO), which includes the academic offerings for the relevant academic year of the PS. The staff member enters details related to the supplementary teaching activity and, more generally, all relevant information (e.g., the Council Resolution reference, the Contract Record, the completed procedures, the duration, etc.).	Teaching registry file in Excel format	Academic Affairs Office	This activity follows a normal distribution, with a mean of 15 min and a deviation of 25%.

Table A9. Post-assignment communications.

Description	Documentation	Responsibility	Process Time
From the teaching registry file, the staff member can extract the table containing the collaborator's and the assignment's data, which are useful for communicating, via email, the completed assignment to two UniBas offices: the Payroll Office and the Infrastructure and ICT Systems Center—IT Services Office (CISIT).	Table containing the collaborator's data and assignment details in Excel format	Table containing the collaborator's data and assignment details in Excel format	This activity follows a normal distribution, with a mean of 15 min and a deviation of 25%.

Table A10. Issuance of a parking pass for campus access.

Description	Documentation	Responsibility	Process Time
If the holder of the ADI position needs to access campus parking with their own car, the person in charge requests, always via email, that the type of car and the license plate be communicated.	Communication via email	Holder of the position/Teaching Office	This activity follows a normal distribution, with a mean of 15 min and a deviation of 25%.
The obtained instructions are forwarded, with an indication of the usage period, always via email, to the Technical Office (UT) of UniBas in order to obtain a pass for access to the outermost parking of the campus and a badge to control the barrier that marks the innermost parking of the same. The UT staff prepares the permit and the card and makes them available for collection by the holder of the position directly at their counter or at the UD.	Pass/permit and badge/card	Technical Office/holder of the position or Teaching Office	This activity follows a normal distribution, with a mean of 10 min and a deviation of 25%.

Table A11. Obligation of the faculty member holding the position at the conclusion of the supplementary teaching activity.

Description	Documentation	Responsibility	Process Time
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 4800 min to simulate the end of the ADI activities.			
At the conclusion of the ADI (supplementary teaching activity), the holder of the position is required (among other obligations outlined in the signed contract) to complete an activity register (Lesson Register), the Word format form of which can be downloaded from the SP website in the section dedicated to internal forms. The holder of the position must insert details related to each lesson (date and content) and complete the final section dedicated to the declaration of hours of activity performed. After completing the register, the holder of the position converts the file from Word to PDF format, digitally signs it using PadES mode, and obtains approval from the faculty member responsible for the course under which the supplementary teaching activity was carried out. Subsequently, the document is sent via email to the UD, using the same method.	Lesson Register in PDF format	Holder of the Position/Teaching Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.

Table A12. Obligations of the Teaching Office at the conclusion of the ADI (supplementary teaching activity).

Description	Documentation	Responsibility	Process Time
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 180 min to simulate waiting for the Director to sign.			
The UD staff member receives the Lesson Register that the holder of the position has sent via email and verifies the correspondence of the hours of activity performed with the hours planned at the time of assigning the position, which is also performed for the purpose of payment processing. They then submit it for approval by the Director of the SP, always through the G-Drive sharing system. Subsequently, the staff member files the register, updating the UD's electronic archive.	Lesson Register in PDF format	Teaching Office/Director of the Primary Structure	This activity follows a normal distribution, with a mean of 20 min and a deviation of 25%.

Table A13. Payment of the position.

Description	Documentation	Responsibility	Process Time
We insert a delay that simulates when the next activity starts due to the availability of resources to work in the next phase of the process; this delay follows an exponential distribution of 180 min for the next activity.			
The UD staff member sends three forms via email to the holder of the position, in which they must do the following: confirm their personal details and provide their bank account information; provide information regarding the type of contribution to be applied during payment processing; and indicate whether they issued an invoice or not. The forms must be digitally signed in PadES mode and returned to the UD via the same method.	Payment forms in PDF format	Holder of the position/Teaching Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.
Upon receiving the payment forms, duly completed and signed, the staff member prepares a request note for the correction/integration of the initial economic booking, based on the contribution position of the person to be paid, and sends it via email to the UAC Accounting Office.	Request note for correction/integration of booking in Word format	Teaching Office/Administration and Accounting Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.
The UAC Accounting staff member performs the accounting operations requested by the UD and completes the request note, adding, in the empty fields, the booking number(s) for the relevant chapters, and sends it back to the UD, always via email.	Request note for correction/integration of booking in Word format	Administration and Accounting Office/Teaching Office	This activity follows a normal distribution, with a mean of 60 min and a deviation of 25%.
The UD staff member, based on the received booking form, prepares the Payment Communication Summary and a payment request note for the position, containing all process details, addressed to the University's Payroll Office (US). The following attachments are included with the note: the Payment Communication Summary; the contract signed with the collaborator; the forms containing data and information necessary for payment; and, if applicable, the positive opinion/authorization of the entity to which the position holder belongs. The payment request note and its attachments are sent to the US after obtaining the digital signature of the SP Director, acquired by sharing it in a dedicated G-Drive folder, and after protocoling via the TITULUS application.	Payment request and attachments in PDF format	Teaching Office/Department Director/Payroll Office	This activity follows a normal distribution, with a mean of 30 min and a deviation of 25%.

Table A14. Note.

Description	Documentation	Responsibility
The UD staff member is responsible for recording the details of each phase of the process (dates, operation numbers, document repertoires, etc.) in the teaching registry file (courses, supplementary teaching activities, tutoring activities, etc.) for the relevant academic year. This ensures that all necessary information can be retrieved by consulting a single file.	Teaching registry in Excel format	Teaching Office

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